Fiscal Year:	FY 2026	**Task Last Updated:**	FY 07/30/2025
PI Name:	Lybrand, Rebecca Ph.D.		
Project Title:	Growing Food on Mars: Determining the Impact of Radiation, Atmospheric Composition, and Rock Substrate on Plant Growth in a Space Rock Garden Experiment		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	95616-5270	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBPS E.9: Plant Studies
Start Date:	12/15/2022	**End Date:**	12/13/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:	Ruby, Anna Maria	**Contact Phone:**	321-867-7065
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/13/2024 per NSSC information (Ed., 10/24/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rodrigues, Jorge Mazza Ph.D. (University of California, Davis) Melotto, Maeli Ph.D. (University of California, Davis) Zaharescu, Dragos Ph.D. (University of California, Davis)		
Grant/Contract No.:	80NSSC23K0371		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	1. SCIENCE GOALS AND OBJECTIVES. As human civilization expands outside of its terrestrial cradle to explore the Moon and Mars, the sources and delivery of nutrients for long-duration missions must be identified and refined. The importance of using local mineral resources for sustaining life, and the bioengineering of such environments remain at the vanguard of sustainable human space exploration. Our overarching goal is to test how ionizing radiation, atmospheric composition, and rock substrate constrain and influence plant growth in deep space exploration, specifically the maintenance of plants in Lunar and Martian environments. This Early Career Investigation (ECI) will produce new publishable findings that integrate how food plants interactively respond to spaceflight stressors (carbon dioxide / CO2 and radiation) and environmental constraints imparted by basalt rocks containing different morphological and elemental arrangements that serve as nutrient sources for plants and microbes in Mars-relevant environments. Once constructed, the Space Rock Garden Experiment (SRGE) will serve as the framework for performing additional plant studies experiments to be proposed through full-ground based proposals and future International Space Station (ISS) flight experiments. We will achieve three objectives: 1) Develop and construct the SRGE, an integrated experimental system capable of controlling the mineral substrate, water, atmospheric and ultraviolet (UV) radiative conditions, and the presence of plants and microbes; 2) Identify how the flux of short wavelength (UV-B) radiation and atmospheric composition influence the rock weathering environment (e.g., nutrient elements compartmentalization), therefore assessing how coupled atmospheric and stellar energy sources influence the formation and habitability of incipient soils; 3) Integrate tomato and N-fixing plant genotypes, arbuscular mycorrhiza, and associated microbiota into the SRGE to assess how rock properties affect the growth and development of plants as viable crops for deep space exploration under increased CO2 and UV-B radiation. 2. METHODOLOGY. We will design, construct, and test the SRGE to simulate plant growth and microbe-mineral interactions under atmospheric and radiation scenarios relevant to Martian landscape. We will assess plant and microbial stress indicators in combination with biogeochemical analyses of major and trace elements in mineral, water and biomass pools. Micro-XCT (X-Ray Computed Tomography) will be used to assess plant root architecture, pore space morphology, and the biogeochemical indicators required to support complex plant life. We performed a pilot study using basalt rock substrates sampled from Mars analog sites in Iceland and confirmed that: i) tomato and lentil plants successfully co-germinated and grew together in basalt rock substrates under ambient conditions; ii) DNA was can be extracted from fresh basalt rock substrates, indicating that the rock materials are capable of hosting microbial life; and iii) a microXCT approach successfully differentiated dense mineral particles, water-filled pores, air-filled pores, and roots from tomato plants grown in the basalt rock substrates.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Rock weathering replenishes nutrients in the environment, shapes geochemical carbon sequestration, and drives physical, chemical, and/or biological processes at a multitude of scales. Globally, lithology type and distribution, climate, and ecosystem activity are predictors of mineral weathering rates and long-term carbon cycling. Our research is examining the intricacies of how contrasting crystalline and glassy basaltic rock substrates weather and release mineral-bound elements that are essential to life using controlled environmental chambers through the Space Rock Garden Experiment (SRGE). We are also examining how morphological and elemental differences in basalt rock composition impacts food plant production under both Earth and Mars atmospheric conditions, which is expanding understanding of how rock substrate composition influences plant growth. Our work is advancing knowledge of sustainable food production using natural (rock-based) fertilization through basalt weathering. Carbon-capture through mineral weathering is another important secondary effect.
Task Progress:	Compiled from final technical report (Ed., 7/30/2025): The biological phase of the experiment was started in Fall 2024. The Science Principal Investigator (Science PI) Zaharescu performed the testing and method development for this component of the project, including the protocols required for the microbial inoculation and plant seed germination steps. To reduce experimental bias, we aimed to reset the microbial community on all substrates. We replaced the original microbial communities with a microbial mix extracted from a granular rock trap previously deployed in an unvegetated glacial till ecosystem in SE Alaska. We added this extract to the granular rock previously subjected to the abiotic treatment in this experiment. We also added a mixture of 16 mycorrhiza species and associated microbes, from a commercial inoculum (Mighty Mycorrhiza, Wildroot Organic Inc., Horseshoe Bay, TX). Root growth vessels were designed and constructed for the plant experiment, as well as the methodology and steps required for plant growth. Plants were grown under 18C average temperature and simulated solar light (12/12 hours day/night cycle, visible spectrum) in the Earth chamber for three months. Watering was performed weekly using presterilized ultrapure water. An undergraduate team member designed and implemented an automatic watering system which greatly improved watering efficiency. After three months, the experimental tubes were extracted from the growth chamber, watered to saturation, and the pore water was collected for chemical analyses. After extraction, plants were gently separated from the substrate in a water-saturated environment to prevent root damage, photographed under polarizing microscope to visualize root-mineral interactions, and morphology, and any attached rock grains removed using a fine brush. Clean plants were then placed on a WinRHIZO scanner bed submerged in water in a transparent tray, and scanned on white background for root architecture analysis. Collected images were processed to increase edge contrast and subsequently analyzed using WinRHIZO software (Regent Instruments Inc.). Soon after scanning, plants were dried at room temperature, separated into below- and above-ground biomass, and weighed. An undergraduate team member was trained on, and subsequently performed, all aspects of plant sampling and root analysis. We are currently completing the root analysis, and the results will be presented in ensuing manuscripts, where significant contributions will result in co-authorship. This project was supported by a multidisciplinary team, and provided training opportunities for 7 undergraduate and graduate students.
Bibliography Type:	Description: (Last Updated: 07/30/2025)

Abstracts for Journals and Proceedings	Matos M, Lybrand RA. "Connecting soils to life on Earth and beyond." Common Ground: Soils Beyond Borders. The Soil Science Society of America (SSSA) 2024 Bouyoucos Summer Conference, San Juan, Puerto Rico, June 10-12, 2024. Abstracts. Common Ground: Soils Beyond Borders. The Soil Science Society of America (SSSA) 2024 Bouyoucos Summer Conference, San Juan, Puerto Rico, June 10-12, 2024. , Jun-2024